VORTEX HACKATHON

- TEAM NAME : TEAM PHOENIX
- NAME OF PROJECT : SMART SURPLUS FOOD COLLECTION & DISTRIBUTION SYSTEM (FWRP)
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INTRODUCTION

Good food should reach people, not bins.

- Hotels, restaurants, events and households may have safe surplus food.

- At the same time, community kitchens, NGOs and people in need may struggle to access meals.

- FWRP is a web-based platform concept that connects food donors with people and organisations who can collect available food.

- The goal is simple: coordinate donations quickly, clearly and responsibly.

DONATE

List surplus food with quantity, type, expiry and pickup address.

DISCOVER

Browse available donations and view pickup details on a map.

CONNECT

Claim a suitable listing and contact the donor for collection.

PROBLEM STATEMENT

The gap is often coordination—not the absence of food.

SURPLUS GOES UNUSED

Food can remain undistributed when donors do not know who can collect it in time.

INFORMATION IS SCATTERED

Donors and community groups may rely on calls or messages to share food details.

PICKUP IS UNCLEAR

Quantity, dietary type, expiry and location need to be communicated clearly.

Problem statement: How can a simple digital platform help people list, find and coordinate safe surplus-food donations before they expire?

OBJECTIVES

What the project aims to achieve

01 Connect donors and receivers

Create one place to list and discover available food.

02 Capture useful details

Record item, quantity, veg/non-veg type, expiry and pickup address.

03 Make pickup easier

Show location information using an embedded map.

04 Support coordination

Allow users to select a listing and claim food.

05 Improve visibility

Provide an admin view of registered users and active donations.

PROPOSED SOLUTION

One platform. Three simple actions.

1 DONATE FOOD

Donor enters contact details, food item, quantity, dietary type, expiry and address.

2 FIND FOOD

Receiver browses available listings and checks the pickup location and expiry.

3 CLAIM & COLLECT

Receiver selects a listing, claims it and contacts the donor to coordinate collection.

Design principle: keep the donation process clear, location-aware and time-sensitive. Food safety and suitability must be confirmed by the people coordinating each pickup.

METHODOLOGY

How the current prototype works

01 Register / Login

Users create an account and sign in.

02 Add donation

Enter food details, expiry and pickup address.

03 Save listing

Donation details are stored in browser local storage.

04 Browse listings

Available food is displayed in a list.

05 View location

An embedded Google Map helps show the address.

06 Claim food

Receiver selects a listing and gets a confirmation message.

Technology shown in the prototype: HTML • CSS • JavaScript • browser localStorage • embedded Google Maps

OUTCOMES & COMMUNITY IMPACT

Expected community value

LESS AVOIDABLE LOSS

Helps make surplus food visible to potential receivers before its expiry time.

FASTER COORDINATION

Keeps quantity, food type, time and pickup address together in one listing.

LOCAL COLLABORATION

Creates a practical connection between donors, NGOs and community members.

Potential impact: better visibility of available food, more organised pickups and stronger community participation.

Note: These are intended outcomes; impact numbers should be reported only after real-world testing.

CHALLENGES & SOLUTIONS

Challenges Faced

- Ensuring food is still safe to consume
- Collecting accurate quantity and expiry details
- Coordinating pickup at the right location
- Prototype data is stored locally in the browser

Steps Taken

- Make expiry and dietary type required fields
- Ask donors to provide clear, truthful details
- Show the pickup address with a map
- Future version: secure shared database and verified accounts

Safety note: the platform coordinates listings; it does not itself certify food quality.

RECOMMENDATIONS & FUTURE SCOP

From a student prototype to a community-ready service

SECURITY

Use a backend database, encrypted communication, secure authentication and role-based access.

SMART MATCHING

Recommend nearby donations based on distance, quantity, dietary needs and expiry time.

COMMUNITY PARTNERS

Onboard NGOs, food banks, caterers and volunteers; define pickup and safety guidelines.

Further enhancements: live notifications • verified donor profiles • donation history • analytics dashboard • mobile-friendly interface

CONCLUSION

A small digital bridge can make a meaningful difference.

FWRP proposes a straightforward way to list surplus food, discover nearby donations and coordinate collection.

The current web prototype demonstrates the core flow: donation entry, listing display, map-based location information and food claiming.

With secure backend services, trusted community partners and clear food-safety procedures, the idea can be developed into a more reliable local redistribution service.

Share food. Strengthen communities. Reduce avoidable loss.

References

1. FWRP student prototype source code (food.html), supplied for this project.

2. MDN Web Docs — HTML, CSS and JavaScript documentation: https://developer.mozilla.org/

3. Google Maps Embed API documentation: https://developers.google.com/maps/documentation/embed

4. Savitribai Phule Pune University / institute CEP presentation format supplied by the student.

Project title: Smart Surplus Food Collection & Distribution System (FWRP)

THANK YOU

Questions & Discussion

!!! Thank you for your time!!!